

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 04: Friday 21 August 2026

(Week 2)

Remark. Q03.7 to Q03.9 are leftover exercises from the Tuesday class in Week 2.

Remark. Q03.7 and Q03.8 are on Section 9.9 of Thomas' Calculus.

Q03.9-Q04.10 are on topics in Sections 11.1-11.4 of Thomas' Calculus.

Q04.11-Q04.15 are on topics in Section 11.5.

Q03.10 (Supplementary Exercise): Find

$$\frac{d}{dx} \int_{x^3}^{\tan x} \cos(t^2) dt.$$

$$\begin{aligned} &= \frac{d}{dx} \left(\int_{x^3}^1 \cos(t^2) dt + \int_1^{\tan x} \cos(t^2) dt \right) \\ &= \frac{d}{dx} \left(- \int_1^{x^3} \cos(t^2) dt + \int_1^{\tan x} \cos(t^2) dt \right) \\ &= -\cos((x^3)^2) \cdot 3x^2 + \cos(\tan^2 x) \cdot \sec^2 x. \end{aligned}$$

6. Power series, Maclaurin Series, Taylor Series (Chapter 9)

Q03.7. (i) Let $f : I \rightarrow \mathbb{R}$ be a function defined on an interval I , and take a point $a \in I$. Let $g(x) = b + c(x - a) + d(x - a)^2$ be a quadratic polynomial, where b, c, d are real constants. Suppose

$$g(a) = f(a), \quad g'(a) = f'(a) \quad \text{and} \quad g''(a) = f''(a).$$

Find b, c, d in terms of $f(a)$, $f'(a)$ and $f''(a)$.

(ii) Now let $a = 1$ and suppose $f(1) = 2$, $f'(1) = 3$ and $f''(1) = 5$. Write down $g(x)$ in this case. Then using $g(x)$ as an approximation of $f(x)$, find an approximate value of $f(1.05)$.

Solution: (i) Differentiating $g(x)$ successively, we get

$$g(x) = b + c(x - a) + d(x - a)^2$$

$$\implies g'(x) = c + 2d(x - a)$$

$$\implies g''(x) = 2d.$$

Then $g(a) = b, g'(a) = c, g''(a) = 2d$

$$g(a) = f(a) \implies b = f(a),$$

$$g'(a) = f'(a) \implies c = f'(a)$$

$$g''(a) = f''(a) \implies 2d = f''(a) \implies d = \frac{f''(a)}{2}.$$

Thus $g(x) = f(a) + f'(a) \cdot (x - a) + \frac{f''(a)}{2}(x - a)^2.$

(ii) Now let $a = 1$ and suppose $f(1) = 2$, $f'(1) = 3$ and $f''(1) = 5$. Write down $g(x)$ in this case. Then using $g(x)$ as an approximation of $f(x)$, find an approximate value of $f(1.05)$.

Solution: Putting $a = 1$, we get

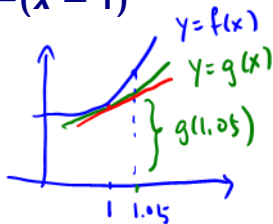
$$\begin{aligned} g(x) &= f(1) + f'(1) \cdot (x - 1) + \frac{f''(1)}{2}(x - 1)^2 \\ &= 2 + 3 \cdot (x - 1) + \frac{5}{2}(x - 1)^2. \end{aligned}$$

Then

$$f(1.05) \approx g(1.05)$$

$$= 2 + 3 \cdot (1.05 - 1) + \frac{5}{2}(1.05 - 1)^2$$

$$= 2.15625.$$



In general, we have (see Section 9.9),

Taylor's Theorem: If f has derivatives of all orders in an open interval I containing a , then for each positive integer n and for each x in I , we have

$$f(x) = \underbrace{f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots}_{g(x)} + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n(x),$$

where

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}$$

for some c between a and x .

If $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$, then we get

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n \quad (*)$$

for all x in I . The right hand side of $(*)$ is called the **Taylor series of $f(x)$** at $x = a$.

When $a = 0$, we get

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \quad (**)$$

for all x in I . The right hand side of $(**)$ is called the **Maclaurin series of $f(x)$** .

For (1), e.g. $x=2$

$$\text{LHS} = \frac{1}{1-2} = -1$$

$$\text{RHS} = \sum_{n=0}^{\infty} 2^n = 1 + 2 + 2^2 + 2^3 + \dots = +\infty.$$

Some standard Maclaurin series:

$$1. \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$1. \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

$$3. e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad |x| < \infty$$

$$4. \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad |x| < \infty$$

$$5. \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \quad |x| < \infty$$

Q03.8. Using the Maclaurin series of e^x given by

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \quad -\infty < x < \infty,$$

find the Taylor series of the function $f(x) = e^{2x-2}$ at $x = 3$.

Solution:

Method I: Compute $f^{(n)}(3)$ for all n .

Taylor series: $\sum_{n=0}^{\infty} \frac{f^{(n)}(3)}{n!} (x-3)^n$.

Method II: (Theorem 21, p. 596 of Thomas' Calculus)

Recall : $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, $|x| < \infty$. (*)

$$\begin{aligned} f(x) &= e^{2x-2} \\ &= e^{2(x-3)+4} \\ &= e^4 \cdot e^{2(x-3)} \end{aligned}$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(3)}{n!} (x-3)^n$$

Replace x
by $2(x-3)$
in (*)

$$(*) = e^4 \cdot \sum_{n=0}^{\infty} \frac{(2(x-3))^n}{n!}$$

$$|2(x-3)| < \infty$$

$$= \sum_{n=0}^{\infty} \frac{e^4 2^n}{n!} (x-3)^n$$

$$|x-3| < \infty$$

↑ Taylor series of $f(x)$ at $x=3$.

Week 2 (Aug 17 - Aug 21) —

Dot products, Determinants, Cross product, Matrices
Equations of planes, Polar coordinates

Read: Thomas' Calculus 15th ed. chapter 11.(1,2,3,4,5),
chapter 10. (3,4)

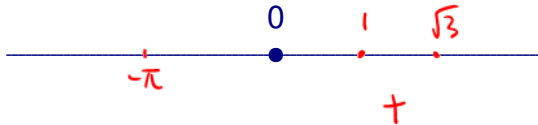
Watch: MIT 18.02 Lec 01 video, notes (Lec 01, 02)

Watch: MIT 18.02 Lec 02 video

Watch: MIT 18.02 Lec 03 video, notes (Lec 03, 04, 05)

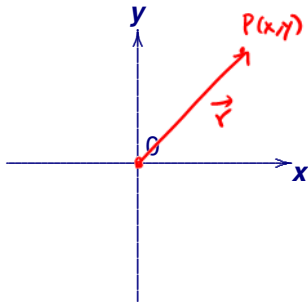
Watch: MIT 18.02 Lec 04 video.

Notation: (i) We denote the set of real numbers by $\mathbb{R}$. Geometrically, $\mathbb{R}$ is represented by the **real line**, where each real number is represented by a point on the real line.



(ii) **The two-dimensional coordinate system** is commonly named as the **xy-plane** or the **Cartesian plane**. Sometimes this is denoted by

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}.$$

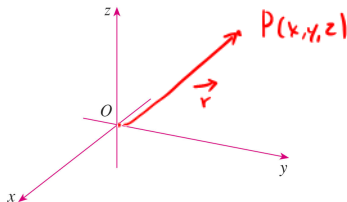


A point (x, y) in $\mathbb{R}^2$ is also represented by its **position vector**

$$\vec{r} = \langle x, y \rangle = xi + yj + \cancel{0k}.$$

(iii) **The three-dimensional coordinate system** is commonly named as the **xyz-space** or simply **space**. Sometimes this is denoted by

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}.$$



A point (x, y, z) in $\mathbb{R}^3$ is also represented by its **position vector**

$$\vec{r} = \langle x, y, z \rangle = xi + yj + zk.$$

(iv) For any positive integer **n** , we also have the **n -dimensional coordinate system** given by

$$\mathbb{R}^n = \{(\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n) : \mathbf{x}_1, \cdots, \mathbf{x}_n \in \mathbb{R}\}.$$

Q03.9. Consider the vectors $\vec{A} = \langle 1, 5, 6 \rangle$ and $\vec{B} = \langle 2, 1, -3 \rangle$.

(i) Find $2\vec{A} - \vec{B}$.

(ii) Find the lengths of $\vec{A}$ and $\vec{B}$.

(iii) Find the unit vector with the same direction as $\vec{B}$.

(iv) Find the dot product $\vec{A} \cdot \vec{B}$.

(v) Hence find the angle between $\vec{A}$ and $\vec{B}$. Give your answer in radians, correct to two decimal places.

Given that

$$\vec{A} = \langle 1, 5, 6 \rangle, \quad \vec{B} = \langle 2, 1, -3 \rangle.$$

(i) Find $2\vec{A} - \vec{B}$.

(ii) Find the lengths of $\vec{A}$ and $\vec{B}$.

Solution: (i)

$$\begin{aligned} 2\vec{A} - \vec{B} &= 2\langle 1, 5, 6 \rangle - \langle 2, 1, -3 \rangle \\ &= \langle 2 \cdot 1, 2 \cdot 5, 2 \cdot 6 \rangle - \langle 2, 1, -3 \rangle \\ &= \langle 0, 9, 12 \rangle. \end{aligned}$$

15

(ii)

$$\begin{aligned} |\vec{A}| &= \|\vec{A}\| = \sqrt{1^2 + 5^2 + 6^2} = \sqrt{1^2 + 5^2 + 6^2} = \sqrt{62}, \\ \|\vec{B}\| &= \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}. \end{aligned}$$

Given that

$$\vec{A} = \langle 1, 5, 6 \rangle, \quad \vec{B} = \langle 2, 1, -3 \rangle.$$

$$\|u\| = 1$$

(iii) Find the unit vector with the same direction as $\vec{B}$.

(iv) Find the dot product $\vec{A} \cdot \vec{B}$.

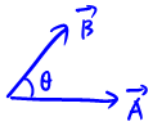
Solution: (iii)

$$u = \frac{\vec{B}}{\|\vec{B}\|} = \frac{\langle 2, 1, -3 \rangle}{\sqrt{14}} = \left\langle \frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, -\frac{3}{\sqrt{14}} \right\rangle.$$

(iv)

$$\vec{A} \cdot \vec{B} = \langle 1, 5, 6 \rangle \cdot \langle 2, 1, -3 \rangle = 1 \cdot 2 + 5 \cdot 1 + 6 \cdot (-3) = -11.$$

$$\vec{A} \cdot \vec{B} = \|\vec{A}\| \cdot \|\vec{B}\| \cdot \cos \theta$$



(v) Hence find the angle between $\vec{A}$ and $\vec{B}$. Give your answer in radians, correct to two decimal places.

Solution: Let θ be the angle between $\vec{A}$ and $\vec{B}$. Then

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{\|\vec{A}\| \cdot \|\vec{B}\|} = \frac{-11}{\sqrt{62} \cdot \sqrt{14}} \approx -0.373$$

$$\Rightarrow \theta \approx \cos^{-1}(-0.373) \approx 1.95.$$

Q04.1. Consider the vectors $\vec{A} = \langle 5, 6 \rangle$ and $\vec{B} = \langle 1, -3 \rangle$ in the xy -plane.

(i) Find $\vec{A} + 3\vec{B}$.

(ii) Find the unit vector with the same direction as $\vec{B}$.

(iii) Find the angle between $\vec{A}$ and $\vec{B}$. Give your answer in degrees, correct to two decimal places.

Q04.2. Consider the unit vector $\mathbf{u} = \frac{1}{\sqrt{6}}\mathbf{i} - \frac{1}{\sqrt{6}}\mathbf{j} + \frac{2}{\sqrt{6}}\mathbf{k}$ and the vector $\vec{A} = 3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$. Find the component of $\vec{A}$ along the direction of $\mathbf{u}$.

Q04.1. Consider the vectors $\vec{A} = \langle 5, 6 \rangle$ and $\vec{B} = \langle 1, -3 \rangle$ in the **xy**-plane.

(i) Find $\vec{A} + 3\vec{B}$.

(ii) Find the unit vector with the same direction as $\vec{B}$.

(iii) Find the angle between $\vec{A}$ and $\vec{B}$. Give your answer in degrees, correct to two decimal places.

Solution: (i)

$$\vec{A} + 3\vec{B} = \langle 5, 6 \rangle + 3\langle 1, -3 \rangle = \langle 5 + 3 \cdot 1, 6 + 3 \cdot (-3) \rangle = \underline{\langle 8, -3 \rangle}.$$

(ii)

$$\mathbf{u} = \frac{\vec{B}}{\|\vec{B}\|} = \frac{\langle 1, -3 \rangle}{\sqrt{1^2 + (-3)^2}} = \frac{\langle 1, -3 \rangle}{\sqrt{10}} = \left\langle \frac{1}{\sqrt{10}}, \frac{-3}{\sqrt{10}} \right\rangle.$$

Given that $\vec{A} = \langle 5, 6 \rangle$ and $\vec{B} = \langle 1, -3 \rangle$.

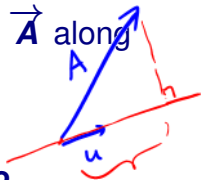
(iii) Find the angle between $\vec{A}$ and $\vec{B}$. Give your answer in degrees, correct to two decimal places.

Solution: Let θ be the angle between $\vec{A}$ and $\vec{B}$. Then

$$\begin{aligned}\cos \theta &= \frac{\vec{A} \cdot \vec{B}}{\|\vec{A}\| \cdot \|\vec{B}\|} = \frac{5 \cdot 1 + 6 \cdot (-3)}{\sqrt{5^2 + 6^2} \cdot \sqrt{1^2 + (-3)^2}} \\ &= \frac{-13}{\sqrt{61} \cdot \sqrt{10}} \approx -0.526 \\ \Rightarrow \theta &\approx \cos^{-1}(-0.526) \approx 121.76^\circ.\end{aligned}$$

Q04.2. Consider the unit vector $\mathbf{u} = \frac{1}{\sqrt{6}}\mathbf{i} - \frac{1}{\sqrt{6}}\mathbf{j} + \frac{2}{\sqrt{6}}\mathbf{k}$ and the vector $\vec{\mathbf{A}} = 3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$. Find the component of $\vec{\mathbf{A}}$ along the direction of $\mathbf{u}$.

Solution: Note that $\|\mathbf{u}\| = 1$. The component of $\vec{\mathbf{A}}$ along the direction of $\mathbf{u}$ is



$$\begin{aligned}\vec{\mathbf{A}} \cdot \mathbf{u} &= (3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}) \cdot \left(\frac{1}{\sqrt{6}}\mathbf{i} - \frac{1}{\sqrt{6}}\mathbf{j} + \frac{2}{\sqrt{6}}\mathbf{k}\right) \\ &= 3 \cdot \frac{1}{\sqrt{6}} + 5 \cdot \left(-\frac{1}{\sqrt{6}}\right) + 7 \cdot \frac{2}{\sqrt{6}} \\ &= \frac{12}{\sqrt{6}} = 2\sqrt{6}.\end{aligned}$$

Q04.3. Consider the two vectors $\vec{A} = \langle 0, -2, -2 \rangle$ and $\vec{B} = \langle 1, 1, -1 \rangle$. Are $\vec{A}$ and $\vec{B}$ orthogonal to each other? Justify your answer.

Q04.4. Consider the vectors $\vec{A} = 6\mathbf{j} + 6\mathbf{k}$ and $\vec{B} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$. Write $\vec{A} = \vec{C} + \vec{D}$ as a sum of two vectors such that $\vec{C}$ is parallel to $\vec{B}$, and $\vec{D}$ is perpendicular to $\vec{B}$.

[Suggestion: First write $\vec{C} = \alpha \vec{B}$ so that $\vec{A} = \alpha \vec{B} + \vec{D}$, where α is a real number. Determine α by using the dot product. Then proceed to get $\vec{C}$ and $\vec{D}$.]

Q04.3. Consider the two vectors $\vec{A} = \langle 0, -2, -2 \rangle$ and $\vec{B} = \langle 1, 1, -1 \rangle$. Are $\vec{A}$ and $\vec{B}$ orthogonal to each other? Justify your answer.

Solution:

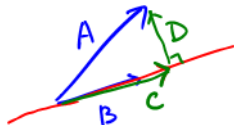
$$\begin{aligned}\vec{A} \cdot \vec{B} &= 0 \cdot 1 + (-2) \cdot 1 + (-2) \cdot (-1) \\ &= 0 - 2 + 2 = 0\end{aligned}$$

$\therefore \vec{A}$ and $\vec{B}$ are orthogonal.

Q04.4. Given $\vec{A} = 6\mathbf{j} + 6\mathbf{k}$ and $\vec{B} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$. Write $\vec{A} = \vec{C} + \vec{D}$ so that $\vec{C} \parallel \vec{B}$ and $\vec{D} \perp \vec{B}$.

Solution: Since $\vec{C} \parallel \vec{B}$, we may write $\vec{C} = \alpha \vec{B}$ for some real number α . Then

$$\vec{A} = \alpha \vec{B} + \vec{D}. \quad (*)$$



Since $\vec{D} \perp \vec{B}$, we $\vec{D} \cdot \vec{B} = 0$. Now taking dot product with $\vec{B}$ on both sides of (*), we have

$$\vec{A} \cdot \vec{B} = \alpha \vec{B} \cdot \vec{B} + \underbrace{\vec{D} \cdot \vec{B}}_{=0} = \alpha \vec{B} \cdot \vec{B} + 0$$

$$\Rightarrow \alpha = \frac{\vec{A} \cdot \vec{B}}{\vec{B} \cdot \vec{B}}$$

$$= \frac{0 \cdot 2 + 6 \cdot (-1) + 6 \cdot (-1)}{2^2 + (-1)^2 + (-1)^2} = \frac{-12}{6} = -2.$$

Hence

$$\vec{C} = -2 \cdot \vec{B} = -2(2\mathbf{i} - \mathbf{j} - \mathbf{k}) = -4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k},$$

$$\vec{D} = \vec{A} - \vec{C}$$

$$= 6\mathbf{j} + 6\mathbf{k} - (-4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$$

$$= 4\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}.$$

Q04.5. Consider the vectors $\vec{A} = \langle 5, 6 \rangle$ and $\vec{B} = \langle 1, -3 \rangle$ in the xy -plane. Find $\det(\vec{A}, \vec{B})$ and $\det(\vec{B}, \vec{A})$.

Q04.6. Consider the vectors $\vec{A} = \langle 1, 2, 3 \rangle$, $\vec{B} = \langle -1, -1, 2 \rangle$ and $\vec{C} = \langle 0, 1, -1 \rangle$ in space. Find $\det(\vec{A}, \vec{B}, \vec{C})$.

Q04.7. Consider the vectors $\vec{A} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\vec{B} = \mathbf{j} - \mathbf{k}$. Find the cross product $\vec{A} \times \vec{B}$.

Q04.5. Given that $\vec{A} = \langle 5, 6 \rangle$ and $\vec{B} = \langle 1, -3 \rangle$. Find $\det(\vec{A}, \vec{B})$ and $\det(\vec{B}, \vec{A})$.

Solution:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\det(\vec{A}, \vec{B}) = \begin{vmatrix} 5 & 6 \\ 1 & -3 \end{vmatrix} = 5 \cdot (-3) - 6 \cdot 1 = -21.$$

$$\det(\vec{B}, \vec{A}) = \begin{vmatrix} 1 & -3 \\ 5 & 6 \end{vmatrix} = 1 \cdot 6 - (-3) \cdot 5 = 21.$$

$|\det(\vec{A}, \vec{B})| = \text{area of // gram spanned by } \vec{A}, \vec{B}.$



Q04.6. Consider the vectors $\vec{A} = \langle 1, 2, 3 \rangle$,
 $\vec{B} = \langle -1, -1, 2 \rangle$ and $\vec{C} = \langle 0, 1, -1 \rangle$ in space. Find
 $\det(\vec{A}, \vec{B}, \vec{C})$. $\mathbb{R}^3$

Solution:

$$\begin{aligned} & \det(\vec{A}, \vec{B}, \vec{C}) \\ &= \begin{vmatrix} 1 & 2 & 3 \\ -1 & -1 & 2 \\ 0 & 1 & -1 \end{vmatrix} \quad + \quad - \quad + \\ &= 1 \cdot \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} - 2 \cdot \begin{vmatrix} -1 & 2 \\ 0 & -1 \end{vmatrix} + 3 \cdot \begin{vmatrix} -1 & -1 \\ 0 & 1 \end{vmatrix} \\ &= 1 \cdot (-1) - 2 \cdot 1 + 3 \cdot (-1) \\ &= -6. \end{aligned}$$

Q04.7. Consider the vectors $\vec{A} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\vec{B} = \mathbf{j} - \mathbf{k}$. Find the cross product $\vec{A} \times \vec{B}$.

Solution:

$$\begin{aligned} & \vec{A} \times \vec{B} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 2 & -1 \\ 0 & 1 & -1 \end{vmatrix} \quad \begin{matrix} + & - & + \end{matrix} \\ &= \mathbf{i} \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 3 & -1 \\ 0 & -1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} \\ &= (-1)\mathbf{i} - (-3)\mathbf{j} + (3)\mathbf{k} \\ &= -\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}. \end{aligned}$$

Q04.8. Consider the points $\mathbf{A} = (5, 2, 0)$, $\mathbf{B} = (2, 6, 1)$, $\mathbf{C} = (2, 4, 7)$ in space. Find the area of the parallelogram with two adjacent sides given by the line segments $\mathbf{AB}$ and $\mathbf{AC}$.

[Suggestion: First write down the vectors $\overrightarrow{\mathbf{AB}}$ and $\overrightarrow{\mathbf{AC}}$. Here $\overrightarrow{\mathbf{AB}}$ denotes the vector with initial point at $\mathbf{A}$ and terminal point at $\mathbf{B}$, etc.]

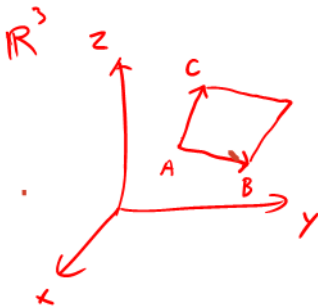
Q04.9. Consider the vectors $\overrightarrow{\mathbf{A}} = \mathbf{i} + 3\mathbf{k}$, $\overrightarrow{\mathbf{B}} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\overrightarrow{\mathbf{C}} = 5\mathbf{i} + 4\mathbf{k}$ in space. Find the volume of the parallelepiped spanned by the above three vectors.

Q04.8. Consider the points $A = (5, 2, 0)$, $B = (2, 6, 1)$, $C = (2, 4, 7)$ in space. Find the area of the parallelogram with two adjacent sides given by the line segments AB and AC .

$$\|\vec{AB} \times \vec{AC}\|$$

= area of the // gram

e.g. $\vec{AB} = \langle -3, 4, 1 \rangle$



Solution: Given that $\mathbf{A} = (5, 2, 0)$, $\mathbf{B} = (2, 6, 1)$,
 $\mathbf{C} = (2, 4, 7)$. Then

$$\overrightarrow{AB} = \langle 2 - 5, 6 - 2, 1 - 0 \rangle = \langle -3, 4, 1 \rangle.$$

$$\overrightarrow{AC} = \langle 2 - 5, 4 - 2, 7 - 0 \rangle = \langle -3, 2, 7 \rangle.$$

Then

$$\begin{aligned}\overrightarrow{AB} \times \overrightarrow{AC} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 4 & 1 \\ -3 & 2 & 7 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} 4 & 1 \\ 2 & 7 \end{vmatrix} - \mathbf{j} \begin{vmatrix} -3 & 1 \\ -3 & 7 \end{vmatrix} + \mathbf{k} \begin{vmatrix} -3 & 4 \\ -3 & 2 \end{vmatrix} \\ &= 26\mathbf{i} + 18\mathbf{j} + 6\mathbf{k}.\end{aligned}$$

Then

Area of the parallelogram

$$\begin{aligned} &= \|\vec{AB} \times \vec{AC}\| = \|\mathbf{26i} + \mathbf{18j} + \mathbf{6k}\| \\ &= \sqrt{\mathbf{26^2} + \mathbf{18^2} + \mathbf{6^2}} \\ &= \sqrt{\mathbf{676} + \mathbf{324} + \mathbf{36}} = \sqrt{\mathbf{1036}} = \mathbf{2\sqrt{259}}. \end{aligned}$$

Q04.9. Consider the vectors $\vec{A} = \mathbf{i} + 3\mathbf{k}$,
 $\vec{B} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\vec{C} = 5\mathbf{i} + 4\mathbf{k}$ in space. Find the
volume of the parallelepiped spanned by the three
vectors.



Solution:

Volume of the parallelepiped = $|\det(\vec{A}, \vec{B}, \vec{C})|$. Now

$$\begin{aligned}\det(\vec{A}, \vec{B}, \vec{C}) &= \begin{vmatrix} 1 & 0 & 3 \\ 2 & 1 & -2 \\ 5 & 0 & 4 \end{vmatrix} \\ &= 1 \begin{vmatrix} 1 & -2 \\ 0 & 4 \end{vmatrix} - 0 \begin{vmatrix} 2 & -2 \\ 5 & 4 \end{vmatrix} + 3 \begin{vmatrix} 2 & 1 \\ 5 & 0 \end{vmatrix} \\ &= 1 \cdot (4) - 0 + 3 \cdot (-5) = -11.\end{aligned}$$

Thus volume of parallelepiped = $|-11| = 11$.

Q04.10. Suppose **A**, **B**, **C** and **D** are four points in space such that $\det(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}) = 0$. Is it true that the 4 points are coplanar (i.e. they all lie on a plane in space)? (You do not need to justify your answer.)

Ans: Yes.



Q04.11. Find an equation of the plane containing the point $(6, 3, 2)$ and perpendicular to the vector $-2\mathbf{i} + 5\mathbf{j} + \mathbf{k}$.

Q04.12. Consider the plane with equation $x + 3y - 7z - 8 = 0$. Find a normal vector of the plane.

Q04.13. Consider the two following two planes:

$$P_1 : x + 3y - 7z - 8 = 0,$$

$$P_2 : -2x - 6y + 14z = 5.$$

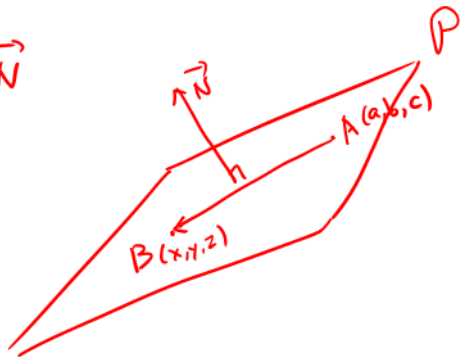
Are the two planes P_1 and P_2 parallel? Justify your answer.

Q04.11. Find an equation of the plane P containing the point $(6, 3, 2)$ and perpendicular to the vector $-2\mathbf{i} + 5\mathbf{j} + \mathbf{k}$.

$$\vec{AB} = \langle x-a, y-b, z-c \rangle \perp \vec{N}$$

$$\Leftrightarrow \langle x-a, y-b, z-c \rangle \cdot \vec{N} = 0$$

↑
Equation of P



Solution: A normal vector of P is $\vec{N} = -2\mathbf{i} + 5\mathbf{j} + \mathbf{k}$.

For any point (x, y, z) in P , since $(6, 3, 2)$ is also in P , the vector $\langle x - 6, y - 3, z - 2 \rangle$ is parallel to P .

Thus an equation of P is

$$\langle x - 6, y - 3, z - 2 \rangle \cdot \vec{N} = 0$$

$$\iff \langle x - 6, y - 3, z - 2 \rangle \cdot (-2\mathbf{i} + 5\mathbf{j} + \mathbf{k}) = 0$$

$$\iff -2 \cdot (x - 6) + 5(y - 3) + 1 \cdot (z - 2) = 0$$

$$\iff \underline{-2x + 5y + z - 5 = 0.}$$